

Pharmaniaga Theophylline Tablet

DESCRIPTION

A white, round, biconvex tablet with Pharmaniaga icon on one side and scoreline on the other.

COMPOSITION

Each tablet contains Theophylline 125 mg.

ACTIONS

Theophylline relaxes smooth muscle and relieves bronchial spasm. It stimulates the myocardium and produces a diminution of venous pressure in congestive heart failure leading to a marked increase in cardiac output. It has a stimulant effect on respiration.

Theophylline is less well absorbed from the gastro-intestinal tract than aminophylline.

At least 60% of Theophylline may be bound to plasma proteins.

Theophylline is excreted in the urine as metabolites, mainly 1, 3-dimethyluric acid and 3-methylxanthine, and about 10% is excreted unchanged.

There are marked variations in Theophylline pharmacokinetics. Plasma half-life for Theophylline ranging from 3 to 9 hours and therapeutic plasma concentrations from 5 to 20 µg per ml have been reported.

Theophylline has been reported to cross the placenta and to diffuse into breast milk.

INDICATIONS

For bronchial asthma and other conditions accompanied by bronchospasm.

ADVERSE REACTIONS 1 SIDE EFFECTS

The side-effects commonly encountered with theophylline are gastro-intestinal irritation and stimulation of the central nervous system. Theophylline may cause nausea, vomiting, abdominal pain, diarrhoea, gastro-intestinal bleeding, insomnia, headache, anxiety, restlessness, dizziness and palpitations.

WARNING / PRECAUTIONS

Pregnancy

Although adequate and well-controlled studies in pregnant women have not done, these medications are used in pregnancy when the risk of treatment is preferable to the risk of placental hypoxaemia from uncontrolled pulmonary disease. The Collaborative Perinatal Project monitored 193 mother-child pairs

exposed to a theophylline during the first trimester and found no evidence of association with teratogenicity.

Theophylline crosses the placenta; cord blood concentrations are approximately equal to the maternal serum concentration. Because of this, higher-than-recommended serum concentrations during pregnancy may result in potentially dangerous serum theophylline and caffeine concentrations in the neonates. Tachycardia, irritability, jitteriness and vomiting have been reported; therefore, neonates of mothers taking these medications during pregnancy should be monitored for signs of theophylline toxicity. Theophylline clearance is reported to be lower in the third trimester, which may necessitate more frequent theophylline serum concentration determinations and possible dosage reduction.

Breast-feeding

Less than 1% of maternal theophylline dose distributes into breast milk; this may cause irritability in the infant.

Paediatrics

Caution is recommended in neonates and children less than 1 year of age; especially in premature neonates and in infant less than 3 months of age with renal function impairment, because theophylline clearance is reduced, resulting in lower dosage requirement. Clearance progressively increases over the first year of life, remains constant during subsequent 9 years, and gradually declines to mean adult values by 16 years of age.

Geriatrics

Caution is recommended in patients older than 60 years of age. Theophylline clearance in these patients is 30% lower than in healthy younger adults. These patients may require adjustment in the dosage or dosing intervals. Severe signs or symptom of toxicity resulting from chronic overdosage are more common in the elderly patients, occurring in 65% of patients 60 years of age or older with serum theophylline concentrations of more than 30 µg per mL.

Tobacco smoking:

Increased clearance of theophylline. Dosage adjustments may be necessary when smoking is started or stopped.

DRUG INTERACTIONS

Alcohol, allopurinol, cimetidine, ciprofloxacin, enoxacin, fluvoxamine, recombinant alpha interferon, clarithromycin, erythromycin, troleandomycin, mexiletine, pentoxifylline, propranolol, tacrine, thiabendazole, ticlopidine:

Decreased theophylline clearance; dosage adjustments may be necessary with concurrent use.

Moricizine, phenytoin, rifampin:

Increased theophylline clearance; dosage adjustments may be necessary with concurrent use.

Beta-blockers including ophthalmic agents:

Concurrent use may result in inhibition of its bronchodilator effect; extreme caution is recommended.

Halothane:

Ventricular arrhythmias have been reported when halothane is used concurrently.

Ketamine:

Concurrent use may lower the seizure threshold.

CONTRAINDICATIONS

The risk-benefit should be considered in patients with acute pulmonary oedema, congestive heart failure, sustained fever, hepatic disease, hypothyroidism, sepsis, seizure disorders and known hypersensitivity to theophylline or ethylenediamine.

DOSAGE AND ADMINISTRATION

Adults	:	125 mg every 6 hours. This can be increased to 250 mg depending on response.
Children	:	7-12 years : 62.5 mg to 125 mg (20-35 kg) depending on (20-35 kg) response.
	:	2-6 years : 5 mg / kg of body-weight every 6 hours.

OVERDOSAGE

Theophylline is associated with a significant potential of toxicity because of its narrow therapeutic range and has caused significant morbidity and mortality. Severe toxicity may not be preceded by milder symptoms.

Symptoms:

- Abdominal pain, continuing or severe.
- Agitation (nervous or restlessness, continuing).
- Confusion or change in behaviour.
- Diarrhoea.
- Haematemesis (dark or bloody vomit).
- Hyperglycaemia
- Hypokalaemia.
- Hypertension (dizziness or light headache)
- Metabolic acidosis.
- Seizures.
- Tachyarrhythmias (fast and irregular heartbeat).
- Tachycardia (fast heartbeat).
- Trembling.

Treatment:

There is no antidote for theophylline overdose.

Treatment is symptomatic and supportive, and includes the following:

- Oral activated charcoal; repeated doses.
- Ipecac syrup should generally be avoided.
- Gastric lavage is generally not necessary if the patient has vomited. Lavage may provide some benefit if performed via a large bore orogastric tube less than 1 hour after a large ingestion.
- Whole bowel irrigation with polyethylene glycol and electrolyte combinations may be of some value if performed early in the treatment.
- Charcoal haemoperfusion is the most effective means of increasing theophylline clearance.
- Seizures must be rapidly controlled. An intravenous benzodiazepine is the drug of choice.
- Serum theophylline concentrations should be monitored continuously until the serum concentrations remains below 20 µg / mL and the patient is asymptomatic.

ROUTE OF ADMINISTRATION

Oral.

STORAGE CONDITIONS

Store below 30°C.
Keep container tightly closed.
Protect from light.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

In blisters of 100 and 500 tablets.

Date of revision: 29th July 2025

PRODUCT	REGISTRATION	HOLDER/
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